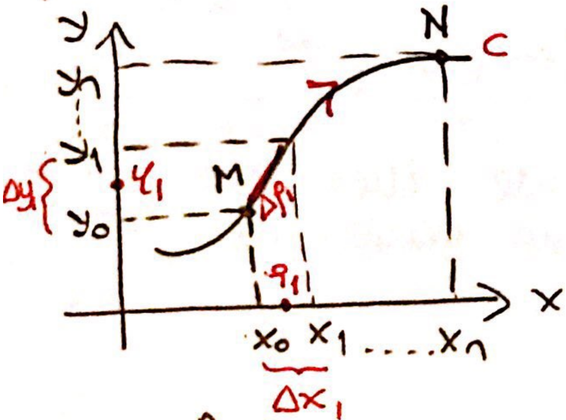
LINE INTEGRAL

Let consider the curve c given by parametric equations $x = x(t)$, $y = y(t)$ and $\overset{\frown}{MN}$ arc on this curve. Let the points M and N correspond to the parameters t_0 and t_n, respectively.

Suppose that the derivatives of the functions $x(t)$, $y(t)$ exist and continuous on $[t_0, t_n]$. Consider two functions $P(x,y)$ and $Q(x,y)$ such that they are continuous and have cont. derivatives at each point of $\overset{\frown}{MN}$ arc. Let cut $\overset{\frown}{MN}$ into parts, arbitrarily, and these points correspond $t_0, t_1, \ldots, t_n$. Choose arbitrary point in each part, $M_i(\xi_i, \eta_i)$



Arc $\Delta \rho_i \rightarrow M_i(\xi_i, \eta_i)$

$$\sum_{i=1}^{n} \left[P(\xi_i, \eta_i)\Delta x_i + Q(\xi_i, \eta_i) \cdot \Delta y_i\right]$$

$n \to \infty$ to have

let $\Delta x_i \to 0$, $\Delta y_i \to 0$ and $\Delta \rho_i \to 0$

$$\lim_{n \to \infty} \sum_{i=1}^{n} \left[P(\xi_i, \eta_i)\Delta x_i + Q(\xi_i, \eta_i)\Delta y_i\right]$$

if this limit exists then it's called the Line integral along $\overset{\frown}{MN}$ arc. It is also defined as;

$$\int_{(M)}^{(N)} P(x,y)dx + Q(x,y)dy \quad \text{or} \quad \int_{\overset{\frown}{MN}} P(x,y)dx + Q(x,y)dy$$

Note: Line integral is defined as the work of the power that affects to a point moving along the curve C.

Similar definitions is also valid for the curves C in space.

$$\int_C P(x,y,z)\,dx + Q(x,y,z)\,dy + R(x,y,z)\,dz$$

Properties :

1) $$\int_{(M)}^{(N)} P(x,y)\,dx + Q(x,y)\,dy = -\int_{(N)}^{(M)} P(x,y)\,dx + Q(x,y)\,dy$$

2) If we have $\widehat{MN} = \widehat{MK} + \widehat{KN}$ then

$$\int_{(M)}^{(N)} P(x,y)\,dx + Q(x,y)\,dy = \int_{(M)}^{(K)} P(x,y)\,dx + Q(x,y)\,dy + \int_{(K)}^{(N)} P(x,y)\,dx + Q(x,y)\,dy$$

3) If the points M and N coincide then the curve C becomes closed curve. In this case; line integral is shown as:

$$\oint P(x,y)\,dx + Q(x,y)\,dy.$$

The positive direction is determined such that inside of the region will be on the left-hand side.

Vector Notation for Line Integrals

Line integral is defined as the work of $\vec{F}$ on a curve C. We can consider $\vec{F}$ as a vector valued function having the components $P(x,y)$, $Q(x,y)$.

Line integral of the vector function $\vec{F}$ along the curve c ;

$$\boxed{\int_{(c)} P(x,y)dx + Q(x,y)dy = \int_{(c)} \vec{F}.d\vec{r}}$$

where $\boxed{\vec{F} = P(x,y)\vec{i} + Q(x,y)\vec{j}}$ and $\boxed{d\vec{r} = dx\vec{i} + dy\vec{j}}$

If the arc length s is considered;

$$\boxed{\int \vec{F}.d\vec{r} = \int\left(\vec{F}.\frac{d\vec{r}}{ds}\right) ds = \int \vec{F}.\vec{T} ds}$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

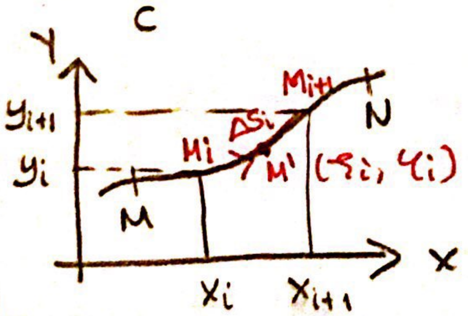
* $\dfrac{d\vec{r}}{dt} = \dfrac{dx}{dt}\vec{i} + \dfrac{dy}{dt}\vec{j} + \dfrac{dz}{dt}\vec{k}$

* $\dfrac{d\vec{r}}{ds} = \vec{T}$: unit tangent vector

* $\dfrac{\frac{d\vec{r}}{dt}}{\left|\frac{d\vec{r}}{dt}\right|} = \dfrac{d\vec{r}}{ds} = \vec{T}$

Calculation of Line Integral

$I = \int_{c} P(x,y)dx + Q(x,y)dy$ and $c : \begin{array}{l} x = g(t) \\ y = h(t) \end{array}$



$$\lim_{n\to\infty} \sum_{i=1}^{n} P(\xi_i, \psi_i)\Delta x_i + Q(\xi_i, \psi_i)\Delta y_i$$

$$\left.\begin{aligned} \lim_{n\to\infty} \sum_{i=1}^{n} P(\xi_i, \psi_i)\Delta x_i &= \int_{\widehat{MN}} P(x,y)dx \\ \lim_{n\to\infty} \sum_{i=1}^{n} Q(\xi_i, \psi_i)\Delta y_i &= \int_{\widehat{MN}} Q(x,y)dy \end{aligned}\right\} \text{limits exist.}$$

These limits are not depending on the type of partition and choice of points.

$x = g(t)$, $y = h(t)$

$\Delta x_i = x_{i+1} - x_i = g(t_{i+1}) - g(t_i)$

Use mean value theorem for $x = g(t)$.

$\Delta x_i = g'(t_i')(t_{i+1} - t_i) = g'(t_i').\Delta t_i \qquad t_i < t_i' < t_{i+1}$

Choose $\xi_i = g(t_i')$, $\psi_i = h(t_i')$. In this case

$$\int_{\widehat{MN}} P(x,y)dx = \lim_{n\to\infty} \sum_{i=1}^{n} P[g(t_i'); h(t_i')].g'(t_i')\Delta t_i$$

As $n \to \infty \Rightarrow \begin{matrix}\Delta x_i \to 0 \\ \Delta y_i \to 0\end{matrix} \Rightarrow \Delta t_i \to 0$ since they are cont. functions.

$$\int_{\widehat{MN}} P(x,y)dx = \int_{t_0}^{t_n} P[g(t), h(t)].g'(t)dt$$

and similarly;

$$\int_{\widehat{MN}} Q(x,y)dy = \int_{t_0}^{t_n} Q[g(t), h(t)].h'(t)dt$$

$$\int_{MN} P(x,y)dx + Q(x,y)dy = \int_{t_0}^{t_n} \{P[g(t),h(t)]g'(t) + Q[g(t),h(t)]h'(t)\} dt$$

A similar calculation is valid for line integral in space.

⊛ If curve c is defined with $y=f(x)$;

$dy = f'(x)dx$

$$\int_C P(x,y)dx + Q(x,y)dy = \int_{a_1}^{a_2} P(x,f(x))dx + Q(x,f(x))f'(x)dx$$

(a_1 and a_2 are on x-axis)

⊛ If curve c is defined with $x=g(y)$;

$$\int_C P(x,y)dx + Q(x,y)dy = \int_{b_1}^{b_2} P(g(y),y)\,g'(y)dy + Q(g(y),y)dy$$

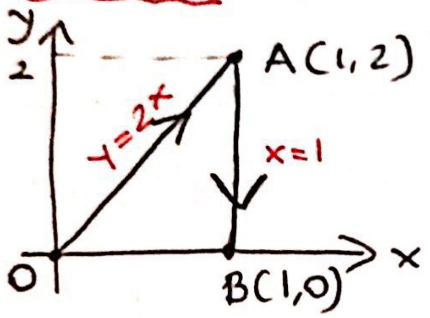
(b_1 and b_2 are on y-axis)

⊛ If parametric equations $x=g(t)$, $y=h(t)$ are given;

$x = g(t) \Rightarrow dx = g'(t)dt$
$y = h(t) \Rightarrow dy = h'(t)dt$

$$\int_C P(x,y)dx + Q(x,y)dy = \int_{t_0(a_0,b_0)}^{t_1(a_1,b_1)} P(g(t),h(t))g'(t)dt + Q(g(t),h(t))h'(t)dt$$

Example: Evaluate $I = \int_C xy\,dx - x^2y^2\,dy$



$C: \overline{OA} + \overline{AB}$

$\quad I_1 + I_2$

* $\overline{OA}$: $y = 2x \Rightarrow dy = 2dx$

$\qquad x = x \Rightarrow dx = dx$

$$I_1 = \int_{\overline{OA}} x.(2x)dx - x^2.4x^2.2dx \Rightarrow$$

$$I_1 = \int_0^1 (2x^2 - 8x^4)dx = \frac{2}{3}x^3 - \frac{8}{5}x^5 \Big|_0^1 = \frac{2}{3} - \frac{8}{5}$$

or: $y = y \Rightarrow dy = dy$

$\qquad x = \frac{y}{2} \Rightarrow dx = \frac{dy}{2}$

$$I_1 = \int_0^2 \frac{y}{2}.y.\frac{dy}{2} - \frac{y^2}{4}.y^2dy = \int_0^2 \left(\frac{y^2}{4} - \frac{y^4}{4}\right)dy$$

$$I_1 = \frac{y^3}{12} - \frac{y^5}{20} \Big|_0^2 = \frac{8}{12} - \frac{32}{20} = \frac{2}{3} - \frac{8}{5}$$

* $\overline{AB}$: $x = 1 \Rightarrow dx = 0$

$\qquad y = y \Rightarrow dy = dy$

$$I_2 = \int_{\overline{AB}} 1.y.0 - 1.y^2dy = \int_2^0 -y^2dy = \frac{y^3}{3}\Big|_0^2 = \frac{8}{3}$$

$$I = I_1 + I_2 = \frac{2}{3} - \frac{8}{5} + \frac{8}{3} = \frac{10}{3} - \frac{8}{5} = \frac{26}{15}$$

3rd way for $\overline{OA}$: $x=t \Rightarrow dx=dt$

$y=2t \Rightarrow dy=2dt$

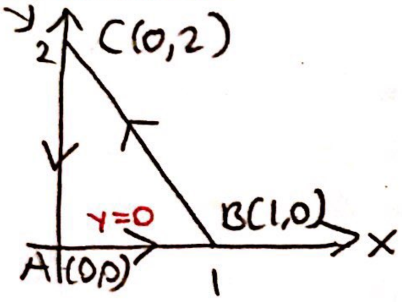
$(x,y) \to t$
$(0,0) \to 0$
$(1,2) \to 1$

$$I_1 = \int_0^1 t.2t.dt - t^2.4t^2.2dt$$

$$I_1 = \int_0^1 2t^2dt - 8t^4dt = \frac{2}{3}t^3 - \frac{8}{5}t^5 \Big|_0^1 = \frac{2}{3} - \frac{8}{5}$$

Example: Evaluate $I = \oint_C (x^3-3y)dx + (x+y^2)dy$

$C : \overline{AB} + \overline{BC} + \overline{CA}$



* $\overline{AB}$: $y=0 \Rightarrow dy=0$

$x=x \Rightarrow dx=dx$

$$I_1 = \int_{\overline{AB}} x^3dx + 0 = \int_0^1 x^3dx = \frac{x^4}{4}\Big|_0^1 = \frac{1}{4}$$

* $\overline{BC}$: $y=2-2x \Rightarrow dy=-2dx$

$x=x \Rightarrow dx=dx$

$$I_2 = \int_{\overline{BC}} x^3 - 3(2-2x)dx + (x+(2-2x)^2)(-2dx)$$

$$I_2 = \int_1^0 [x^3-6+6x-2x-2(4-8x+4x^2)]dx$$

$$I_2 = \int_1^0 (x^3-6+4x-8+16x-8x^2)dx$$

$$I_2 = \int_1^0 (x^3-8x^2+20x-14)dx = \frac{x^4}{4} - \frac{8}{3}x^3 + 10x^2 - 14x\Big|_1^0$$

$$I_2 = 0 - \frac{1}{4} + \frac{8}{3} - 10 + 14 = -\frac{1}{4} + \frac{8}{3} + 4$$

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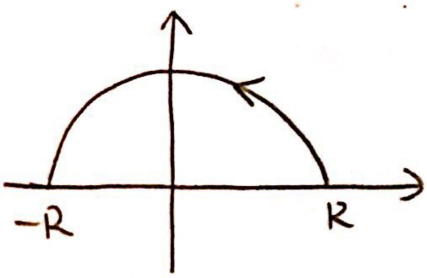
* $\overline{CA}$: $x = 0 \Rightarrow dx = 0$
$y = y \Rightarrow dy = dy$

$$I_3 = \int_{\overline{CA}} -3y \cdot 0 + (0 + y^2)\,dy = \int_2^0 y^2\,dy = \frac{y^3}{3}\Big|_2^0 = 0 - \frac{8}{3}$$

$$I = I_1 + I_2 + I_3$$

$$I = \cancel{\frac{1}{4}} + \cancel{\frac{8}{3}} - \cancel{\frac{1}{4}} + 4 - \cancel{\frac{8}{3}} = 4 \,//$$

Example: Evaluate $I = \int_C x\,dy - y\,dx$



C: $\left.\begin{array}{l} x = R\cos t \\ y = R\sin t \end{array}\right\}$ parametric equations.

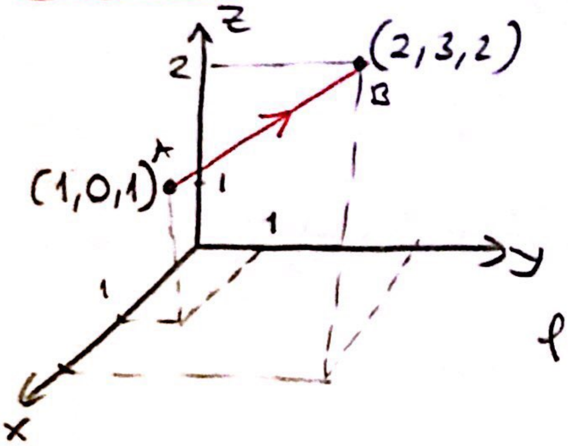
$$\left.\begin{array}{ll} x = R\cos t & \Rightarrow dx = -R\sin t\,dt \\ y = R\sin t & \Rightarrow dy = R\cos t\,dt \end{array}\right\} \quad t: 0 \to \pi$$

$$I = \int_0^{\pi} R\cos t \cdot R\cos t\,dt + R\sin t \cdot R\sin t\,dt$$

$$I = \int_0^{\pi} (R^2\cos^2 t + R^2\sin^2 t)\,dt = \int_0^{\pi} R^2\,dt = R^2 \cdot t\Big|_0^{\pi}$$

$$I = R^2\pi$$

Example: Evaluate $I = \int_{(1,0,1)}^{(2,3,2)} x^2dx - xzdy + y^2dz$



$\overrightarrow{AB} = (1,3,1)$ direction of line ℓ

$\ell:$ $\left.\begin{aligned} x &= 1+t \\ y &= 0+3t \\ z &= 1+t \end{aligned}\right\}$ parametric equations of line d.

$x = t+1 \Rightarrow dx = dt$
$y = 3t \Rightarrow dy = 3dt$
$z = t+1 \Rightarrow dz = dt$

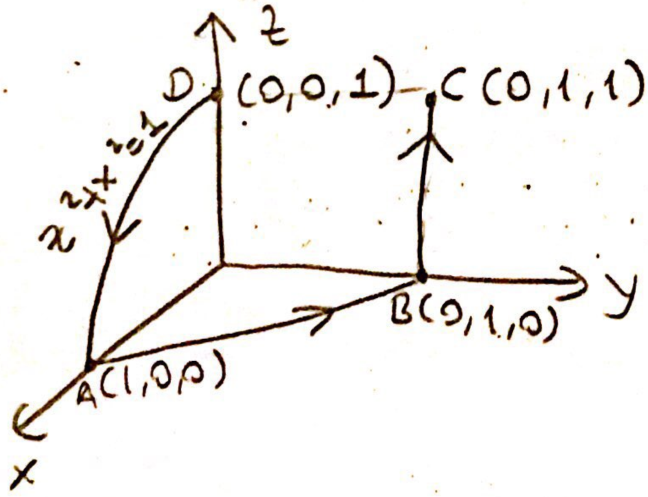
$A(1,0,1) \rightarrow t=0$
$B(2,3,2) \rightarrow t=1$

$$I = \int_0^1 (t+1)^2dt - (t+1)(t+1)3dt + 9t^2dt$$

$$I = \int_0^1 (-2\underbrace{(t+1)^2}_{t^2+2t+1} + 9t^2)dt = \int_0^1 (-2t^2-4t-2+9t^2)dt$$

$$I = \int_0^1 (7t^2-4t-2)dt = \frac{7}{3}t^3 - 2t^2 - 2t \Big|_0^1 = \frac{7}{3} - 4 = -\frac{5}{3}$$

Example: $\vec{F} = xz\vec{i} + x\vec{j} - yz\vec{k}$ $\quad \int_C \vec{F}d\vec{r} = ?$



$C: \widehat{DA} + \overline{AB} + \overline{BC}$

* $\widehat{DA}$: $\left.\begin{array}{l} x = x \\ z = \sqrt{1-x^2} \\ y = 0 \end{array}\right\}$ $\left.\begin{array}{l} dx = dx \\ dz = \dfrac{-x}{\sqrt{1-x^2}}dx \\ dy = 0 \end{array}\right\}$ $d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$

$$I_1 = \int_{\widehat{DA}} \vec{F}.d\vec{r} = \int_{\widehat{DA}} (xz\vec{i} + x\vec{j} - yz\vec{k})(dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$I_1 = \int_{\widehat{DA}} xz dx + x \underset{0}{dy} - yz dz$$

$$I_1 = \int_0^1 x.\sqrt{1-x^2} dx - 0.dz = \int_0^1 x\sqrt{1-x^2}dx$$

$1-x^2 = t$

$-2x dx = dt$

$$I_1 = -\frac{1}{2}\int_1^0 \sqrt{t} dt = -\frac{1}{\cancel{2}}.t^{3/2}.\frac{\cancel{2}}{3}\Big|_1^0 = -\frac{1}{3}(0-1) = \frac{1}{3}$$

* $\overline{AB}$: $\left.\begin{array}{l} x = x \\ y = 1-x \\ z = 0 \end{array}\right\}$ $\begin{array}{l} dx = dx \\ dy = -dx \\ dz = 0 \end{array}$

$$I_2 = \int_{\overline{AB}} xz dx + x dy - yz dz = \int_1^0 x.(-dx) = -\frac{x^2}{2}\Big|_1^0 = \frac{1}{2}$$

* $\overline{BC}$: $\left.\begin{array}{l} x = 0 \\ y = 1 \\ z = z \end{array}\right\}$ $\begin{array}{l} dx = 0 \\ dy = 0 \\ dz = dz \end{array}$

$$I_3 = \int_{\overline{BC}} xz dx + x dy - yz dz = \int_0^1 -1.z.dz = -\frac{z^2}{2}\Big|_0^1 = -\frac{1}{2}$$

$$I = I_1 + I_2 + I_3 = \frac{1}{3} + \frac{1}{2} - \frac{1}{2} = \frac{1}{3}$$